

Finding Hidden Structure in Complex Systems

From Differential Geometry to Quantitative Finance

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Research Advisor: Professor Trung "Average" Phan

Research Highlight: Geometric ML for Financial Regime Detection

Problem. Market regime shifts (2008 GFC, COVID-19, SVB collapse) cause catastrophic portfolio losses. Traditional detection methods require labeled historical crisis data, making them blind to novel crisis types. Unsupervised alternatives lack the sensitivity to distinguish true regime changes from noise.

Approach. I developed a geometric machine learning framework — QCML (Quantum Cognitive Machine Learning) — a quantum-inspired framework that lifts financial time-series into a projective Hilbert space. While the name references quantum mechanics, the detection signals are rooted in differential geometry and spectral theory applied to classical financial data. In this space, market states become points on a manifold. From this representation, I extract three novel geometric observables:

- **Berry phase rate** — measures the rate of geometric rotation in state space
- **Quantum Fisher information** — quantifies the distinguishability between nearby market states
- **Multi-lag fidelity** — measures state overlap decay across multiple time horizons

These observables serve as unsupervised regime-change indicators: they spike when the geometric structure of market returns changes, without requiring any crisis labels.

Validation. Benchmarked against 17 detection methods (including Random Forest, HMM, BOCPD, Isolation Forest, CUSUM) across 16 historical crises on 5 ETFs (SPY, QQQ, IWM, EFA, DIA), spanning 1997–2024. All comparisons use walk-forward validation with bootstrap confidence intervals and permutation tests.

Additional Experience

Quanta Ventures Fund — Production Quant Research

5-sleeve architecture aggregating 133 ML signals (XGBoost, Ridge, mean-reversion) with covariance-based risk budgeting. Walk-forward validation with 30-day embargo and 11-test validation suite. **Sharpe 2.92 | Calmar 5.02** (true out-of-sample, 2022–2025).

Adaptive Vol Arb Platform — Systems Engineering

Built delta-neutral variance-risk-premium strategy with optimized C++ FFT pricer (pybind11). **40x latency reduction** (150ms to <5ms). 38,460 lines across Python (20,760), C++ (1,130), React (4,678) with 24 test files + 24 validation scripts and CI/CD. **Sharpe 1.32** (in-sample).

Key Results.

d = 0.56

Best geometric detector (MLF)
Matches RF rank — no labels
needed

86%

Walk-forward detection rate
9-day median delay, 12 crises

- Multi-Lag Fidelity achieves Cohen's $d = 0.56$ (median across 12 crises). MLF ties RF ($d = 0.96$) on Friedman rank (both 4.42) — **without any crisis labels**
- Wins 5/12 crises RF misses: SVB ($d = 0.94$ vs 0.12), Repo (0.91 vs 0.02)
- Geometric signals are **faster** to compute than RF (0.26s vs 1.07s per window)
- Statistically significant (Friedman $\chi^2 = 49.2$, $p < 10^{-6}$)
- Granger causality: geometric signals predict absolute returns ($F = 9.5$, $p = 0.0004$)

Publication. 25-page paper with 3 theorems, 1 proposition, and 33 peer-reviewed references. Available upon request.

Tools. Python, PyTorch, NumPy/SciPy, scikit-learn, Optuna, Polygon.io API, LaTeX

So What? This work demonstrates:

- Novel ML feature design from first principles (differential geometry)
- Honest benchmarking: unsupervised method ties supervised RF on rank
- Production validation: walk-forward 86% detection, 9-day delay
- Independent end-to-end research execution (solo author, sophomore year)

Honors & Qualifications

- Incoming Quant Research Intern, Steadfast Financial LP (Summer 2026)
- Carpe Diem Endowed Scholar, Pitzer College
- Harvey Mudd Putnam Math Competition Team
- CS Teaching Assistant — mentoring 60+ students
- Graduate-level coursework: Mathematical Finance, Stochastic Calculus
- Languages: Python, C++, SQL, TypeScript, Mathematica, LaTeX
- Methods: Stochastic calculus, Monte Carlo, Bayesian inference, options pricing, game theory

Seeking: Quantitative Research Internship — Summer 2026/2027

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